

CSA0404 – OPERATING SYSTEMS – LAB PROGRAMS

EXPERIMENT 17

Question: Illustrate the deadlock avoidance concept by simulating Banker's algorithm with C.

Program:

```
1 #include <stdio.h>
2
3 #define MAX_PROCESSES 5
4 #define MAX_RESOURCES 3
5
6 int is_safe();
7
8 int available[MAX_RESOURCES] = {3, 3, 2};
9 int maximum[MAX_PROCESSES][MAX_RESOURCES] = {
10     {7, 5, 3},
11     {3, 2, 2},
12     {9, 0, 2},
13     {2, 2, 2},
14     {4, 3, 3}
15 };
16 int allocation[MAX_PROCESSES][MAX_RESOURCES] = {
17     {0, 1, 0},
18     {2, 0, 0},
19     {3, 0, 2},
20     {2, 1, 1},
21     {0, 0, 2}
22 };
23
24 int request_resources(int process_num, int request[]) {
25     for (int i = 0; i < MAX_RESOURCES; i++) {
26         if (request[i] > available[i] || request[i] > maximum[process_num][i])
27             return 0;
28     }
29 }
```

Enter process number (0 to 4): 1
Enter resource request (e.g., 0 1 0): 0 1 0
Request granted.

...Program finished with exit code 0
Press ENTER to exit console.